WORKED SOLUTION

STEP II 2013, Question 9

Mechanics • Friction in Equilibrium

STEP II 2013, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q9

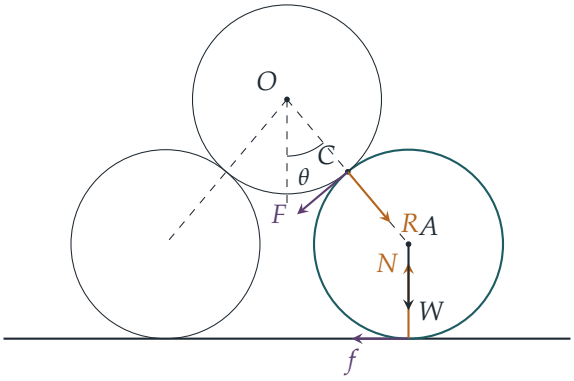
The key idea

Symmetry does most of the work. Treating all three discs as one system gives the ground reactions straight away. Then take moments about the centre of a bottom disc: the friction at the ground and the friction from the top disc both act along tangents, so they have the same lever arm (the radius) and must be equal in size. Once you know that, resolving forces on the bottom disc and on the top disc gives two equations for the two unknowns.

Solution

We model the discs as identical uniform rigid discs of radius r, each of weight W, all in one vertical plane. Let O be the centre of the top disc and A the centre of the right-hand bottom disc, so $OA = 2r$ and OA makes angle θ with the downward vertical. The contact point C between these two discs is the midpoint of OA.

By the left–right symmetry, the forces on the left-hand bottom disc mirror those on the right-hand one, so we only need to work with the right-hand disc. At C the right-hand disc feels a normal reaction R along OA (pushing it away from O) and a friction force F along the common tangent. At the ground it feels a normal reaction N and a friction force f.



The diagram shows the forces on the right-hand bottom disc. By Newton's third law the top disc feels R and F in the opposite directions at C, so friction pushes the top disc upwards and outwards along the tangent.

Part (i)

Take the three discs together as one system. The forces between discs are internal and cancel, so the only external forces are the three weights and the forces from the ground. The two ground reactions are equal by symmetry, and the frictional forces at the ground are horizontal, so resolving vertically gives

$$2N = 3W \quad \Longrightarrow \quad N = \tfrac{3}{2}W,$$

as required.

Part (ii)

> **Ask yourself** There seem to be three unknowns, R, F and f, on the bottom disc. Where does a third equation come from?

From moments. Take moments about A for the right-hand bottom disc. The weight W, the normal reaction N and the normal reaction R all pass through A, so they have no moment. The friction f acts along the tangent at the lowest point and F acts along the tangent at C, so each has lever arm r. They must turn the disc in opposite senses, so

$$fr = Fr \quad \Longrightarrow \quad f = F.$$

(This also fixes the directions in the diagram. The disc is pushed outwards, so f points inwards, and then F on the bottom disc must point down and inwards along the tangent to balance its moment.)

The unit vector from O to A is $(\sin\theta, -\cos\theta)$, and the tangent direction of F on the bottom disc is $(-\cos\theta, -\sin\theta)$. Resolving horizontally for the bottom disc:

$$R\sin\theta - F\cos\theta - f = 0 \quad \Longrightarrow \quad R\sin\theta = F(1+\cos\theta),$$

using $f = F$.

For the top disc, the two contacts give equal upward components, each $R\cos\theta + F\sin\theta$. Resolving vertically:

$$2(R\cos\theta + F\sin\theta) = W.$$

Substituting $F = \dfrac{R\sin\theta}{1+\cos\theta}$:

$$W = 2R\left(\cos\theta + \frac{\sin^2\theta}{1+\cos\theta}\right) = 2R\cdot\frac{\cos\theta+\cos^2\theta+\sin^2\theta}{1+\cos\theta} = 2R\cdot\frac{1+\cos\theta}{1+\cos\theta} = 2R.$$

So the normal reaction between two discs in contact is

$$R = \tfrac{1}{2}W,$$

and it does not depend on θ. Then

$$F = \frac{R\sin\theta}{1+\cos\theta} = \frac{W\sin\theta}{2(1+\cos\theta)},$$

which is the required frictional force.

As a consistency check, resolving vertically for the bottom disc gives $N = W + R\cos\theta + F\sin\theta = W + \frac{1}{2}W = \frac{3}{2}W$, agreeing with part (i).

Part (iii)

Using the half-angle formulae $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$ and $1+\cos\theta = 2\cos^2\frac{\theta}{2}$,

$$\frac{F}{R} = \frac{\sin\theta}{1+\cos\theta} = \tan\tfrac{\theta}{2}.$$

Equilibrium needs friction not to exceed its limiting value at every contact. Between the discs this means $F \le \mu R$, that is,

$$\tan\tfrac{\theta}{2} \le \mu.$$

(At the ground we need $f \le \mu N$, i.e. $\frac{1}{2}W\tan\frac{\theta}{2} \le \frac{3}{2}\mu W$. This is weaker, so the contact between the discs is the one that matters.)

Ask yourself What stops θ being as small as we like, which would make $\tan\frac{\theta}{2}$ tiny?

The bottom discs are not in contact. Their centres are a distance $2\cdot 2r\sin\theta = 4r\sin\theta$ apart, and this must exceed $2r$, so $\sin\theta > \frac{1}{2}$. Since $0 < \theta < \frac{\pi}{2}$, this means $\theta > \frac{\pi}{6}$, so $\frac{\theta}{2} > \frac{\pi}{12}$. As tan is increasing on $(0, \frac{\pi}{2})$,

$$\tan\tfrac{\theta}{2} > \tan\tfrac{\pi}{12} = \tan(45^\circ - 30^\circ) = \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2} = 2-\sqrt{3}.$$

So equilibrium requires $\mu \ge \tan\frac{\theta}{2} > 2-\sqrt{3}$. If $\mu < 2-\sqrt{3}$, no allowed value of θ satisfies this, and equilibrium is impossible.

Checks

- When $\theta \to 0$ the top disc sits directly on the others and $F \to 0$: no friction is needed, as you'd expect.
- For any sample angle, such as $\theta = 1$ radian, substituting $R = \frac{1}{2}W$, $F = f = \frac{1}{2}W\tan\frac{1}{2}$ and $N = \frac{3}{2}W$ makes both force components and the moment balance for each disc.
- $2 - \sqrt{3} \approx 0.2679$, which matches $\tan 15^\circ \approx 0.2679$. A numerical scan over θ just above 30° confirms equilibrium is impossible for $\mu = 0.2679$ and possible for $\mu = 0.268$.

Common slips

- Assuming the friction between the discs is zero or forgetting it altogether. Without it the bottom disc's moment equation forces $f = 0$, and nothing would stop the bottom discs sliding apart.
- Missing the condition that the bottom discs do not touch. It is the only thing that bounds θ from below, and so it is where $2 - \sqrt{3}$ comes from.
- Getting the direction of F wrong. Use the moment equation about A to fix it rather than guessing.

Beyond STEP

This is a statically determinate problem only because of symmetry and the moment equation; in many piles of rough bodies the friction forces cannot be found from equilibrium alone, and engineers need to model how the bodies deform. The neat result $F/R = \tan\frac{\theta}{2}$ is an example of the friction-angle idea: equilibrium is possible exactly when the total contact force lies within an angle $\arctan\mu$ of the normal. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body statics carefully at first-year level.

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